TechMart Practical - Show & Explain Guide

For Muhammad Hussain - read this once, then open dashboard.html and show it step by step

WHAT YOU ARE SHOWING (the whole story in 1 line)

You took a raw, messy customer file, cleaned and verified it, entered it into **MySQL** and **MongoDB**, and produced sales reports from both. Total sales figure everywhere is the same: **Rs 195,700** (cancelled orders excluded). Best city: **Lahore**. Best product: **Monitor 24 inch**.

HOW TO SHOW IT - 5 STEPS (about 3-4 minutes)

- **1. Open dashboard.html:** This is the summary page. Point to the top cards: 12 customers, 16 orders, 8 products, total sales.
- **2. Show the data cleaning section:** Raw file had 15 rows: 2 customers entered twice, cities in mixed case (lahore / KARACHI), phones in 3 formats, 1 missing email, 1 row with no name. You cleaned it to 12 correct customers.
- **3. Show the Excel file (TechMart_Data_Practical.xlsx):** Sheets: Raw_Data_Unclean, Data_Cleaning_Log, Customers_Clean, Products, Orders, Sales_Report. This proves the Excel / data-entry side.
- **4. Show 01_mysql_practical.sql:** Scroll slowly: CREATE TABLE with primary/foreign keys, INSERT data, checking queries (duplicates, missing email), then JOIN + GROUP BY reports.
- **5. Show 02_mongodb_practical.js:** Same data as documents - orders inside the customer document, no JOIN. Show the aggregation pipeline: $unwind, $match, $group, $sort. Same total result as MySQL.

If they ask you to RUN it live:
MySQL: open MySQL Workbench, open 01_mysql_practical.sql, press the lightning (Execute) button, then run report query 5c. MongoDB: open mongosh / Compass, paste 02_mongodb_practical.js sections one by one. Practice this once alone BEFORE the call so there is no panic.

WHAT TO SAY (simple English, you can use these lines)

> *"This is my practical project. I worked with customer and order data of a small store."*
>
> *"First I cleaned the raw data in Excel - I removed duplicates, fixed the city names and phone formats, and flagged one missing email instead of inventing it."*
>
> *"Then I entered the data in MySQL in three tables - customers, products and orders - connected with foreign keys."*
>
> *"I also did the same data in MongoDB, where each customer is one document and his orders are inside it."*
>
> *"From both databases I made the same reports, like sales by city, and both gave the same total, so I know the data is correct."*

QUESTIONS THEY MAY ASK - SHORT ANSWERS

- **Q: Difference between MySQL and MongoDB?**
  A: MySQL stores data in tables with a fixed schema and joins tables with keys. MongoDB stores flexible JSON-like documents; related data can be kept inside one document, so often no join is needed.
- **Q: What is a primary key / foreign key?**
  A: Primary key uniquely identifies each row - it can never repeat. Foreign key is a column that links to another table's primary key, like customer_id in orders links to customers.
- **Q: How do you find duplicate records?**
  A: In MySQL: GROUP BY email HAVING COUNT(*) greater than 1. In MongoDB: $group on email and count. In Excel: Remove Duplicates / Conditional Formatting.
- **Q: How do you check missing or wrong data?**
  A: Check for NULL or empty values, check quantities and prices are not zero or negative, and compare

totals between two systems or reports - if totals match, data is verified.

- **Q: What is aggregation in MongoDB?**
  A: It is like GROUP BY in SQL - a pipeline of stages: $match filters, $unwind opens arrays, $group totals, $sort orders the result.
- **Q: What is a JOIN?**
  A: Join combines rows from two or more tables using a common column, for example orders with customers to show the customer name with each order.
- **Q: Why did you exclude cancelled orders from sales?**
  A: Cancelled orders are not real sales. Including them would give a wrong, higher total - checking the status is part of data verification.
- **Q: Your Excel skills?**
  A: Entering and formatting data, sorting, filtering, TRIM and basic formulas, Remove Duplicates, and making summary reports - all shown in the workbook.

ONE HONEST LINE TO REMEMBER

If they ask where you did this work, say: *"It is my own practical / practice project to show my MySQL, MongoDB and data entry skills."* That is true, it sounds confident, and this project is strong enough that you do not need to claim it was a client job.